3.6.4 Different types of waves

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 7			
Waves may be either transverse or longitudinal. In a transverse wave the oscillations are perpendicular to the direction of energy transfer. The ripples on a water surface are an example of a transverse wave. In a longitudinal wave the oscillations are parallel to the direction of energy transfer. Longitudinal waves show areas of compression and rarefaction. Sound waves travelling through air are longitudinal.	Students should be able to identify whether a wave is longitudinal or transverse from a given diagram.	6.6.1.1	4.1.4.1
Outcome 8			
Waves are described by their amplitude, wavelength and frequency. The amplitude of a wave is the maximum displacement of a point on a wave away from its undisturbed position. The wavelength of a wave is the distance from a point on one wave to the equivalent point on the adjacent wave. Students may be required to use the equation: wave speed (m/s) = frequency (Hz) x wavelength (m) The frequency of a wave is the number of waves passing a point each second.	Students should be able to identify wavelength and amplitude on a given diagram.	6.6.1.2	4.1.4.2